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Node-Centric Meta Structure Search in Heterogeneous Graphs

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***Abstract*—Heterogeneous graphs are increasingly used to represent complex real-world scenarios with diverse entities and interactions by meta structures. Recently, the search of meta structures is combined with graph neural architecture search to automatically extract the semantic knowledge for various tasks in heterogeneous graphs. However, prior research primarily focuses on identifying meta structures that are universally applicable across all nodes in a graph, neglecting the variations in meta structure selection that arise from the unique features and topology of individual nodes. To address this challenge, we introduce a *Node-Centric* approach to search *Meta Structures* in heterogeneous graphs (NC-MS for short). NC-MS implements a node level method that discover meaningful meta structures tailored to each node, capturing subtle differences in meta structure choices between nodes and providing nuanced identification. Additionally, NC-MS utilizes an efficient and differentiable network to enhance operational efficiency. Empirical studies across three real-world datasets validate the superiority of NC-MS, demonstrating its ability to outperform existing models in heterogeneous graph neural networks.**

***Index Terms*—Meta structure, Heterogeneous graphs, Node-Centric, Neural architecture search**

I. Introduction

Heterogeneous Graph Neural Networks (HGNNs) [1]–[4] have emerged as powerful analysis tools for studying heterogeneous graphs. The goal of representing heterogeneous graphs is acquiring vector representations capable of efficiently preserving the relationships and semantic information inherent in nodes and edges from the original graph within a lower-dimensional space. HGNNs have found extensive applications in various real-world domains, including natural language processing [5]–[7], recommendation systems [8]–[10], molecular structure prediction [11]–[13], and anomaly detection [14].

From the view of how structural and semantic information is captured [15], existing heterogeneous graph representation methods can be divided into three categories: link-based, meta path-based, and meta structure-based methods. Link-based methods utilize first-order neighbor information, such as AspEm [16] and HeGAN [17], which mainly analyzes the local structural features and do not take higher-order linkages into consideration. Meta path-based methods address this limitation by considering sequences of relations between node types, such as metapath2vec [18] and HAN [19]. To better capture the complex topological and semantic information, meta structure-based methods add cross-layer topological information, such as Mg2vec [20], DHNE [21] and HLWP [22].

Although meta structure-based methods can describe higher-order node similarities and help to capture complicated relationships, their complicated structures make the manual designation difficult. To address this issue, several neural architecture search (NAS) methods [24]–[28] have been developed to automatically search for meta structures in heterogeneous graphs. For example, GEMS [29] employs evolutionary algorithms, while DiffMG [23] and PMMM [30] utilize differentiable architecture search (DARTS [31]) to find a common meta structure applicable to all nodes in a heterogeneous graph.

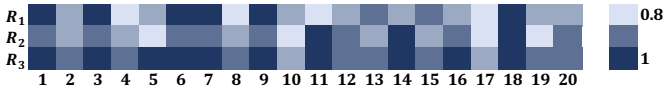


Fig. 1: Nodes exhibit diverse preferences in selecting meta structures in the same heterogeneous graph. The x-axis shows the node id, the y-axis shows the different representations based on meta paths or structures, and the shades of color represent the correct prediction rate for 20 independent experiments. The data source is 20 random nodes in ACM (details in [23]). R_1 represents the meta structure searched by DiffMG, R_2 represents the meta structure searched by PMMM, R_3 represents the meta structures searched by NC-MS (ours).

However, these NAS methods overlook the varying preferences for meta structures among nodes. To validate this, we examined the impact of diverse meta structures on the node prediction task. As depicted in Figure 1, it is evident that distinct nodes exhibit varied preferences in choosing meta structures in the same heterogeneous graph, and the performance difference caused by different meta structures is obvious. Consequently, when representing nodes and aggregating their neighbors, it is crucial to search tailored meta structures for different nodes in a heterogeneous graph.

To this end, this paper proposes a ***N**ode-**C**entric **M**eta **S**tructure* approach in heterogeneous graphs (**NC-MS** for short), which can automatically design the meta structures for each node based on their features. NC-MS is a tailored method that focuses on utilizing node-specific features to enhance the representation and analysis of each individual node in the same heterogeneous graph. Specifically, NC-MS leverages node-specific features to improve the representation and analysis of individual nodes, capturing the nuanced semantic contexts. Different from previous methods, NC-MS guides the different semantic neighbors' selection through the node's features and the overall network structure, allowing it to skillfully design a unique meta structure for each node based on task characteristics and node features. Moreover, NC-MS employs an efficient and differentiable approach to search for meta structures. This mechanism enhances search efficiency and enables NC-MS to outperform in web service prediction tasks on HGNNs. Last but not least, NC-MS builds upon DiffMG and shares the same search space, resulting in a 1%-3% improvement over DiffMG without requiring a larger search space. What's more, NC-MS shares almost the same time consumption with existing models (for details in table III), such as DiffGM and PMMM.

Our main contributions are summarized as follows:

- We introduce NC-MS, a novel node-centric approach, to auto-

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matically search for unique meta structures tailored to individual nodes, guided by node's specific features and structure.
- We design a novel search algorithm specifically designed to identify node-centric meta structures within heterogeneous graphs, enabling a more targeted and effective exploration.
- We conduct a several assessment for NC-MS on various real-world heterogeneous graph datasets. The outcomes reveal that NC-MS outperform the existing baselines.

II. Preliminaries

A. Definitions

Definition 1. ***Heterogeneous Graph.*** *A heterogeneous graph is defined as a graph $\mathcal{G}=(\mathcal{V},\mathcal{E},\mathcal{T},\mathcal{R})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges, $\mathcal{T}$ and $\mathcal{R}$ are the set of node's type and edge's type. In particular, a heterogeneous graph $\mathcal{G}$ requires satisfying $|\mathcal{T}|+|\mathcal{R}|>2$.*

We denote the set of adjacency matrices $\mathcal{A}$ as:

$$\mathcal{A}=\{\mathbf{A}_r\}_{r\in\mathcal{R}}, \tag{1}$$

where $\mathbf{A}_r$ is the adjacency matrix of edge type r, and $\mathcal{R}$ is the edge type set that belongs to the heterogeneous graph $\mathcal{G}$. Based on Equation (1), given set of task t related edge types as $\mathcal{R}_t$, and $\mathcal{R}_t\subset\mathcal{R}$, we define the set of task t related adjacency matrices as:

$$\bar{\mathcal{A}}(t)=\{\mathbf{A}_r\}_{r\in\mathcal{R}_t}. \tag{2}$$

Since the task-related node type t is fixed in the given task, for the sake of simplicity, ***we omit the t in $\bar{\mathcal{A}}(t)$ and denote $\bar{\mathcal{A}}(t)$ by in $\bar{\mathcal{A}}$*** in subsequent sections of this paper.

Definition 2. ***Heterogeneous Graph Schema.*** *Given a heterogeneous graph $\mathcal{G}=(\mathcal{V},\mathcal{E},\mathcal{T},\mathcal{R})$, its schema $\mathcal{S}_\mathcal{G}$ is a directed graph defined over set of node's type $\mathcal{T}$ and edge's type $\mathcal{R}$, i.e., $\mathcal{S}_\mathcal{G}=(\mathcal{T},\mathcal{R})$.*

Definition 3. ***Meta Path.*** *A meta path mp is a composite relation defined on $\mathcal{S}_\mathcal{G}$, which consists of multiple edge types. Formally, $mp:=\{t_i\xrightarrow{r_{i+1}}t_{i+1}\}_{0\le i<l}$, where $t_0,\cdots,t_l\in\mathcal{T}$ and $r_1,\cdots,r_l\in\mathcal{R}$, and l is the length of the meta path, describing the layers in which messages are propagated between nodes.*

Definition 4. ***Meta Structure***. *A meta structure ms is a directed cyclic graph with a single target node n_t (with out-degree 0), defined on a heterogeneous graph schema $\mathcal{S}_\mathcal{G}=(\mathcal{T},\mathcal{R})$. Formally, $ms:=\left\{t_i\xrightarrow{r_{ij}}t_j\right\}_{0\le i<j\le l}$, where l denotes the maximum depth (or number of hops) in the message passing process of the meta structure, $t_0,\cdots,t_l\in\mathcal{T}$, $r_{ij}\in\mathcal{R}$, and $t_l=n_t$.*

Definition 5. ***Node-Centric Meta Structure.*** *The set of node-centric meta structures $\mathcal{PMS}$ belongs to the target nodes set $\mathcal{V}$ in heterogeneous graph tasks. Each target node v has its own set of tailored meta structures pms_v, and the complete collection of these structures is represented as $\mathcal{PMS}=\{pms_v\}_{v\in\mathcal{V}}$.*

Remark 1. *Different target nodes can have different node-centric meta structures, which is different from previous research that all nodes share the same meta structure.*

B. Problem setup

Our goal is to search suitable node-centric meta structures $\{pms_v^*\}_{v\in V}$ for each node v in the heterogeneous graph $\mathcal{G}$. We achieve this by minimizing the loss ℓ as follows:

$$\bigotimes_{v\in\mathcal{V}}pms_v^*=\arg\min_{\bigotimes_{v\in\mathcal{V}}pms_v\in\bigotimes_{v\in\mathcal{V}}\mathcal{PMS}_v}\sum_{v\in\mathcal{V}}\ell(\tilde{y}_v,y_v), \tag{3}$$

$$\tilde{y}_v=f(\mathcal{G},pms_v), \tag{4}$$

where $\bigotimes_{v\in\mathcal{V}}pms_v^*$ denotes the set of node-centric meta structures for nodes set $\mathcal{V}$, $\mathcal{PMS}_v$ denotes the set of all candidate node-centric meta structures for node v. y_v and $\tilde{y}_v$ denote the true value and predicted value of node v. $f(\mathcal{G},pms_v)$ denotes the calculation of $\tilde{y}_v$ using function f based on pms_v in the heterogeneous graph $\mathcal{G}$.

III. Methodology

In this section, we first introduce the node-centric meta structure based message passing in Section III-A. Next, we introduce the candidate generation of node-centric meta structure in Section III-B. Then, we introduce the node-centric meta structure candidate selection based on policy gradient in Section III-C. In the last part, we discuss the model training and prediction for specific tasks in Section III-D. The illustration of NC-MS shows in Figure 2.

A. Node-Centric Meta Structure Based Message Passing

Message-passing based HGNNs learn node representations by aggregating information from nodes' neighbors via the message-passing scheme (edge type r). This process is expressed as follows:

$$\boldsymbol{H}^{(i)}=\mathrm{F}\left(\boldsymbol{H}^{(i-1)},\mathbf{A}_r\right), \tag{5}$$

where $\boldsymbol{H}^{(i)}$ represents the node embeddings in the i-th layer, $\mathbf{A}_r$ denotes the adjacency matrix with edge type r, and F is the aggregate function combining information from neighboring nodes and edges. When $i=0$, $\boldsymbol{H}^{(0)}=\boldsymbol{X}$, where $\boldsymbol{X}$ represents the node features. Common choices for F include summation, mean, or more advanced mechanisms such as attention, depending on the specific HGNN.

In meta structure based message passing, $\boldsymbol{H}^{(i)}$ represents the aggregation in layer i-th from different source layers (j-th) and Equation (5) omits the layer information. So we define $\boldsymbol{H}^{(i)}$ in meta structure based message passing as:

$$\boldsymbol{H}^{(i)}=\sum_{j=0}^{i-1}\mathrm{F}\left(\boldsymbol{H}^{(j)},\mathbf{A}_r^{(j,i)}\right),0<i\le l, \tag{6}$$

where $\mathbf{A}_r^{(j,i)}$ is a matrix, represents message passing between j-th and i-th layer with edge type t, $\mathbf{A}_r^{(j,i)}\in\mathcal{A}^{(j,i)}$. We define $\mathcal{A}^{(j,i)}$ as:

$$\mathcal{A}^{(j,i)}=\begin{cases}\mathcal{A}\cup\{\boldsymbol{I}\} & j=i-1 \text{ and } i<l\\ \bar{\mathcal{A}}\cup\{\boldsymbol{I}\} & j=i-1 \text{ and } i=l\\ \mathcal{A}\cup\{\boldsymbol{I}\}\cup\{\boldsymbol{O}\} & j<i-1 \text{ and } i<l\\ \bar{\mathcal{A}}\cup\{\boldsymbol{I}\}\cup\{\boldsymbol{O}\} & j<i-1 \text{ and } i=l\end{cases}, \tag{7}$$

where $\mathcal{A}$ is the set of adjacency matrices, $\bar{\mathcal{A}}$ is the set of task t focused adjacency matrices defined in Equation (2). $\boldsymbol{I}$ is the identity matrix and $\boldsymbol{O}$ is the zeros matrix . The matrices $\boldsymbol{I}$ and $\boldsymbol{O}$ allow the number of actual message passing layers to be flexible.

B. Node-Centric Meta Structure Generation

In this part, we design two functions to generate node-centric meta structures and make the algorithm efficient.

1) ***Projection Function****:* We use projection function to transformer the nodes' features from original space to target space:

$$\boldsymbol{X}'\leftarrow\boldsymbol{X},(\boldsymbol{X}\in\mathbb{R}^{|\mathcal{V}|\times d},\boldsymbol{X}'\in\mathbb{R}^{|\mathcal{V}|\times d'}), \tag{8}$$

where d and d' are the nodes' dimension in original and target space, $\boldsymbol{X}$ and $\boldsymbol{X}'$ are the nodes' features in original and target space, $|\mathcal{V}|$ is nodes' number. We define a projection function F_p to project the node features $\boldsymbol{X}$, and $\boldsymbol{X}'=\mathrm{F}_p(\boldsymbol{X})$. Because we transform the node features just according to the graph structure and its node features, so this process is unsupervised. Specifically, we use a method called

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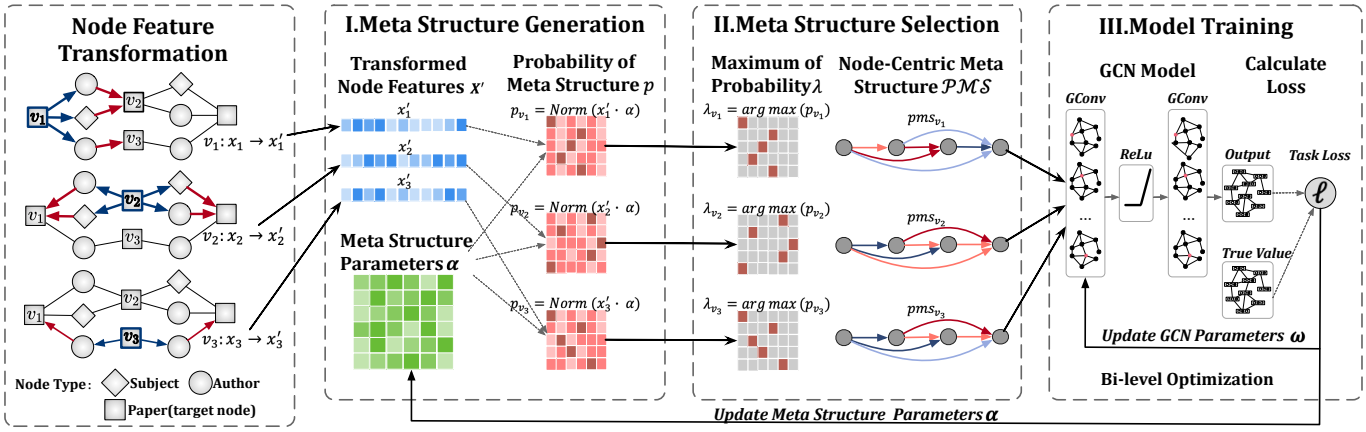


Fig. 2: The concepts and process illustrations of NC-MS. In part I, NC-MS generates the node-centric meta structure for each node and gets the probability of selecting the meta structure for each node (Section III-B). In part II, NC-MS selects each node's most likely meta structure (Section III-C). In part III, NC-MS trains a GCN model by bi-level optimization and predicts the results of tasks (Section III-D).

Algorithm 1 NC-MS Algorithm

Require: heterogeneous graph $\mathcal{G}$, set of task focused adjacency matrices $\bar{\mathcal{A}}$.

1: Initialization: meta structure parameters matrix $\boldsymbol{\alpha}$, GNN parameters ω;
2: Prepossess node features $\boldsymbol{X}' \leftarrow \boldsymbol{X}$; ▷ see Eq.(8)
3: **while** not convergence **do**
4: Compute the MS probability of the node $\bar{p}_a^{(j,i)}$; ▷ see Eq.(11)
5: Select the most suitable meta structure ▷ see Eq.(13)
6: Update ω^*, $\nabla_{\omega^*}\ell_{\text{tra}}(\omega,\alpha)$; ▷ see Eq.(15)
7: Update α, $\nabla_{\alpha}\ell_{\text{val}}(\omega^*(\alpha),\alpha)$; ▷ see Eq.(14)
8: **end while**
9: Export meta structures of nodes and GNN model;

return: meta structures of nodes and GNN model.

uniform manifold approximation and projection [32] as the projection function to generate target dimension of nodes.

*2) **Custom Function**:* We propose a custom function that transforms the converted node features $\boldsymbol{X}'$, into a matrix set representing the probability of each node's candidate meta structures. Specifically, $\alpha^{(j,i)}$ is the set of parameters between the j-th and i-th layers. The item $\alpha_r^{j,i}$ represents the parameter of edge type r. $\mathcal{R}^{(j,i)}$ is the set of candidate edge types between i-th and j-th layers, which corresponding to $\mathcal{A}^{(j,i)}$ in eq. (7) and $r \in \mathcal{R}^{(j,i)}$.

$p_r^{(j,i)}$ represents the probability of edge type r when passing between j-th and i-th layers, which based on the converted node features $\boldsymbol{X}'$ and parameter $\alpha_r^{(j,i)}$. It is calculated as follows:

$$p_r^{(j,i)} = \boldsymbol{X}' \cdot \frac{\lambda\,\alpha_r^{(j,i)}}{\left\|\alpha_r^{(j,i)}\right\|_2}, \tag{9}$$

where λ is the hyper-parameter of the *L2* normalization. We adopt the *L2* normalization on $\alpha_r^{(j,i)}$ to prevent model over fitting and enhance model generalization ability. Then we use soft-max to handle $p_r^{(j,i)}$:

$$\hat{p}_r^{(j,i)} = \frac{e^{p_r^{(j,i)}}}{\sum e^{p_r^{(j,i)}}}. \tag{10}$$

This process can avoided the local maximum and minimum problem.

C. Node-Centric Meta Structure Selection

To improve search efficiency of node-centric meta structures, we transform the discrete candidate node-centric meta structures into continuous. Specifically, we achieve continuity in the selection of discrete edge types through the above formula:

$$\bar{p}_r^{(j,i)} = \frac{\hat{p}_r^{(j,i)}}{\sum_{r' \in \mathcal{R}^{(j,i)}} \hat{p}_{r'}^{(j,i)}}, \tag{11}$$

where $\bar{p}_r^{(j,i)}$ represents the relaxed probability. This enables the training for all potential edge types through back propagation, which can improve search efficiency. Then we select the highest probability meta structure from the set $\{\bar{p}_a^{(j,i)}\}_{a \in \mathcal{R}^{(j,i)}}$ for node v:

$$\lambda_v^{(j,i)} = \arg\max_{a \in \mathcal{R}^{(j,i)}} (\bar{p}_a^{(j,i)}), v \in \mathcal{V}, \tag{12}$$

where $\bar{p}_a^{(j,i)}$ is calculated based on Equation (11).

We aggregate the maximum probability $\lambda_v^{(j,i)}$ across all layers, and denote this set as λ_v. Then we get the meta structure of node v, represented as pms_v, corresponds to λ_v:

$$\underset{v \in \mathcal{V}}{\otimes} pms_v^* \leftarrow \underset{v \in \mathcal{V}}{\otimes} \lambda_v^*. \tag{13}$$

D. Model Training

We train the meta structure parameters α and the GNN parameters ω. Specifically, we update ω by minimizing ℓ_{tra} with the fixed α in the training set, and update α by minimizing ℓ_{val} with a fixed ω in the validation set. This iterative optimization process can be formulated as a bi-level optimization problem [33]:

$$\min_{\alpha} \ell_{\text{val}}(\omega^*(\alpha),\alpha), \tag{14}$$

$$s.t. \quad \omega^* = \arg\min_{\omega} \ell_{\text{tra}}(\omega,\alpha), \tag{15}$$

where ℓ_{val} denotes the loss on the validation set, and ℓ_{tra} denotes the loss on the training set. After convergence, we get model parameters ω and meta structure parameters α to predict the downstream tasks.

IV. Experiment

A. Experimental Setup

We evaluate NC-MS [1] on node classification tasks. We use three widely used real-world datasets: ACM, DBLP, and IMDB (details

[1]Our code is released at https://github.com/Febeoro/NC-MS.

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TABLE I: Node classification performance comparison. The best and second-best results are shown in **bold** and underlined.

	ACM		DBLP		IMDB	
	macro-f1	micro-f1	macro-f1	micro-f1	macro-f1	micro-f1
GCN	92.56±0.20	91.22±0.64	90.46±0.41	91.47±0.36	55.19±0.99	54.63±1.03
GAT	92.50±0.23	92.86±0.29	93.92±0.28	93.39±0.30	53.37±1.27	53.64±1.43
HAN	91.20±0.25	91.57±0.38	92.13±0.26	92.05±0.62	55.09±0.67	56.37±0.78
MAGNN	91.15±0.19	91.06±0.33	92.81±0.30	92.58±0.21	56.44±0.63	57.60±0.72
GTN	92.62±0.17	91.78±0.59	93.98±0.32	94.01±0.28	59.68±0.72	59.40±1.44
HGT	91.83±0.23	91.12±0.23	93.67±0.22	93.49±0.25	59.35±0.79	59.42±0.74
ie-HGCN	92.73±0.17	91.92±0.43	94.01±0.25	94.09±0.12	59.78±0.87	59.83±1.23
GraphMSE	92.58±0.50	92.54±0.49	94.08±0.14	94.44±0.13	57.60±2.13	62.37±1.03
DiffMG	92.65±0.15	92.21±0.31	94.45±0.15	94.85±0.27	61.04±0.56	61.81±1.59
PMMM	92.76±0.14	92.65± 0.12	94.69±0.10	95.40±0.10	61.69±0.40	64.31±0.77
NC-MS(ours)	**93.24±0.04**	**92.86±0.09**	**94.81±0.02**	**95.50±0.05**	**62.11±0.11**	**65.54±0.13**

TABLE II: Redundancy study between NC-MS and random MS. The superior results are emphasized in **bold**.

	ACM		DBLP		IMDB	
	ma-f1	mi-f1	ma-f1	mi-f1	ma-f1	mi-f1
Random(avg)	87.54	87.23	81.15	82.12	49.82	50.40
NC-MS(avg)	**93.20**	**92.83**	**94.81**	**95.49**	**61.95**	**64.84**
Random(best)	92.90	92.85	85.34	86.10	54.59	56.01
NC-MS(best)	**93.54**	**93.46**	**94.87**	**95.52**	**62.31**	**65.84**

TABLE III: Time cost compared with automated machine learning algorithms on HINs, measured in GPU minutes.

Method	Amazon	Yelp	Douban
GEMS	400.5	750.7	1001
DiffMG	0.6	0.9	1.3
PMMM	0.8	1.1	1.6
NC-MS (ours)	0.7	1.0	1.5

in [23], [34]). We compare NC-MS with several popular GNNs, including: two homogeneous GNNs: GCN [35] and GAT [36], two heterogeneous GNNs required manually designed meta paths: HAN [19] and MAGNN [37], three heterogeneous GNNs which use attention-based meta paths: GTN [38], HGT [39] and ie-HGCN [40], and four auto machine learning methods GraphMSE [41], GEMS [29], DiffMG [23] and PMMM [30]. We perform three separate runs of the search algorithm, each initiated with distinct random seeds. The final meta structure for evaluation is derived from the run that attains the highest validation performance. We align our parameters with those of the pivotal baseline, DiffMG, for meaningful comparison. During the search stage, parameter α is trained using the Adam optimizer with a learning rate of $3e^{-4}$. In contrast, parameter ω is trained with Adam using a learning rate of 0.005 and a weight decay of 0.001. Across all datasets, we consistently set $L = 4$, aligning with the meta path length derived from GTN. All experiments use a single RTX 2080 Ti GPU with 11GB memory. For each method, we repeat the process for ten runs with different random training seeds and report the average score and standard deviation.

B. Node Classification Results

The node classification results are shown in Table I. Firstly, compared to homogeneous GNNs (GCN and GAT), most heterogeneous GNNs demonstrate better performance, indicating that ignoring node semantic information is not advisable. Interestingly, homogeneous GNNs perform better than manually designed heterogeneous GNNs (such as HAN and MAGNN). This highlights the limitations of human-designed rules for extracting task-specific semantic insights. Secondly, manual-design heterogeneous GNNs (HAN and MAGNN) usually display comparatively sub-optimal results when contrasted with attention-based models, including GTN, HGT, and ie-HGCN. This observation underscores the criticality of adaptation to varying neighbors, which is essentially a node-centric process. Thirdly, NC-MS exhibits a markedly competitive performance compared with attention-based heterogeneous GNNs (GTN, HGT, and ie-HGCN) based on meta paths. This is because NC-MS can aggregate message neighbors more extensively and expand the message acceptance domain by using meta structures. At the same time, NC-MS can provide more intuitive interpret-ability than attention-based heterogeneous GNNs. Fourth, three auto-machine learning methods (GraphMSE, DiffMG, and PMMM) outperform heterogeneous GNNs employing meta paths, demonstrating the power of NAS and the advantages of meta structures. NC-MS performs better than the above three methods, indicating the advantage of node-centric meta structure. Notably, NC-MS consistently attains superior performance across all three datasets. This robustly underscores the implementation of node-centric meta structures in heterogeneous graphs.

C. Redundancy Study and Efficiency

We conduct experiments to verify the effectiveness of meta structures selected by NC-MS and compare them with randomly initialized meta structures. To ensure the robustness, we conducted ten trials for each of the fifty different random seeds and computed the average and maximum values for each evaluation metric. This evaluation is carried out on three distinct datasets, using dataset partitioning consistent with the methodology described in Section IV-B. We employ both macro-f1 and micro-f1 scores as performance metrics. As depicted in table II, results demonstrate the consequent of node-centric meta structure search over the random initialization approach.

We compare the time cost of NC-MS against existing works that employ NAS for heterogeneous GNNs . The time costs reported in table III encompass both the training and search stages. GEMS proves to be time-consuming as it requires individual evaluation of each candidate model. For DiffMG and PMMM, we present the overall model training costs for ten runs, as detailed in the table III. Compared to DiffMG and PMMM, NC-MS exhibits similar search efficiency and performs better.

V. Conclusion

In this paper, we propose a node-centric meta structure approach in heterogeneous graphs, named NC-MS. NC-MS employs a node-level method to identify customized meta structures that are specifically tailored to each node, thereby capturing the subtle variations in meta structure selection among nodes and ensuring nuanced identification. Furthermore, NC-MS utilizes an efficient and differentiable network to improve operational efficiency. Empirical studies conduct on node classification tasks and three real-world datasets suggest the effectiveness of NC-MS, indicating its potential to enhance performance over existing models in heterogeneous graph neural networks.

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